

Nafiul Islam Naim

B.Sc. (Honours), 4th-Year Student
Department of Applied Mathematics
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🎓 Education

Bachelor of Science Oct 2022 – Expected Dec 2026
Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh
CGPA: 3.00/4.00 (through 3rd year)

Undergraduate Project : “ Principal Component Analysis and Its Applications ” — Developed a PCA and machine learning framework for heatwave analysis and prediction using ERA5 reanalysis climate data across Bangladesh’s eight administrative divisions .

Supervisor: [Shuchi Chaki](#)

🔬 Research Interests

My research interests focus on **Data Science** , **Climate Science** , **Deep Learning** , **Computer Vision** and **Multimodal AI** with a particular emphasis on video understanding, anomaly detection, object detection , explainable AI, and climate-data-driven modelling . I am looking to work with complex data and use computational and AI techniques to find solutions to real world problems, specifically finding efficient, resource-limited, and interpretable solutions.

📖 Research Project (Ongoing)

Sub-Seasonal Heatwave Onset Prediction across Bangladesh’s Divisions Using Regional Atmospheric and Global Teleconnection Predictors with Multiple Lead Times

- Created a novel 1991–2025 multi-region climate dataset covering five heatwave-prone regions of Bangladesh—Dhaka, Rangpur, Rajshahi, Khulna, and the Chittagong Hill Tracts (RBK), integrating 35 atmospheric, land-surface, and global teleconnection predictors for sub-seasonal heatwave onset prediction.
- Built an explainable machine-learning framework using mRMR feature selection, classification ML models (Logistic Regression, Random Forest, SVM), and SHAP analysis to identify and interpret the most influential climate drivers of sub-seasonal heatwave onset.

</> Projects

Principal Component Analysis and Its Applications

[🔗 Code](#)

- Developed an interpretable PCA-based machine learning framework to predict annual Heat Wave Days (HWDs) across Bangladesh’s eight administrative divisions using ERA5 atmospheric data .
- Applied mRMR feature selection and PCA dimensionality reduction to identify informative atmospheric predictors and address high-dimensional, correlated climate variables.
- Compared multiple regression models and used SHAP explainability to identify the dominant atmospheric drivers influencing heat-wave occurrence across different regions of Bangladesh.

Tools: Python, Pandas, NumPy, Scikit-learn, PCA, mRMR, SHAP, ERA5

📄 Standardized Tests

I plan to take the **International English Language Testing System (IELTS)** and the **Graduate Record Examination (GRE)** soon.

Technical skills

Programming	Python, C, C++, Dart , MATLAB
Data Analysis & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
ML & Deep Learning	PyTorch, Keras, SHAP Analysis
Climate & Geospatial Analysis	ERA5/ERA5-Land, NetCDF, Xarray, GeoPandas, GIS, Spatial Analysis, Grid-Based Analysis, Remote Sensing (NDVI, NDWI), Google Earth Engine(GEE), Climate Anomaly & Teleconnection Analysis(SST, ENSO, IOD, MJO, OLR)
Computer Vision	Ultralytics YOLO, OpenCV, Roboflow
Web & Application Development Tools	HTML, CSS, Flutter Git, GitHub, VS Code, PyCharm, LaTeX, Google Colab, Jupyter Notebook, Canva

Referees

- **Shuchi Chaki**

Graduate Teaching Assistant

North Carolina State University

Email: shuchi@du.ac.bd

Lecturer (on study leave)

Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh

- **Kawser Ahammed**

PhD Student (fifth year)

Department of Electrical and Computer Engineering

Old Dominion University, Virginia, USA

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Assistant Professor (on study leave)

Department of Electrical and Electronic Engineering (EEE),

[Jatiya Kabi Kazi Nazrul Islam University, Trishal, Mymensingh, Bangladesh](#)